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BDA400 – Data Science Tools and Techniques

Assignment 5 – Technical Analysis using R, Development Phase

GitHub Repository:

<https://github.com/neeraj-19-98/markdown-buddy-neeraj/tree/main>

1. Introduction

This assignment focuses on implementing and testing technical analysis indicators using R. The main purpose is to create reusable R functions for commonly used technical analysis calculations and verify that the functions work correctly with sample data.

The implementation includes Simple Moving Average (SMA), Exponential Moving Average (EMA), Moving Average Convergence Divergence (MACD), Standard Deviation, Linear Regression, Relative Strength Index (RSI), Stochastic RSI, Crossover, and Crossunder.

Each indicator was implemented as a separate R function and tested using sample data. The testing results were checked in RStudio to make sure the functions produced the expected outputs.

2. Simple Moving Average (SMA)

The Simple Moving Average calculates the average value over a specified number of periods. It helps smooth the data and identify the general direction of a trend.

Formula:

$$\text{SMA} = \text{Sum of values over the period} / \text{Number of periods}$$

Implementation: sma.R

Test period: 3

3. Exponential Moving Average (EMA)

The Exponential Moving Average gives more importance to recent values compared with the Simple Moving Average. It is useful for identifying trends while responding more quickly to recent changes in the data.

Formula:

$$\text{EMA} = (\text{Current Value} \times \text{Smoothing Factor}) + (\text{Previous EMA} \times (1 - \text{Smoothing Factor}))$$

Implementation: ema.R

Test period: 3

Test Result: The EMA function was successfully executed in RStudio and produced the expected numerical output.

4. Moving Average Convergence Divergence (MACD)

MACD is a trend-following indicator based on the difference between a fast EMA and a slow EMA. A signal line is then calculated from the MACD line, and the histogram shows the difference between the MACD line and signal line.

Implementation: macd.R

Test periods: Fast = 3, Slow = 5, Signal = 2

Test Result: The MACD function was successfully executed in RStudio and returned MACD, signal, and histogram values.

5. Standard Deviation

Standard Deviation measures how much the values vary from their average. A higher value indicates greater variation, while a lower value indicates that the values are closer to the average.

Implementation: stdev.R

Test period: 3

Test Result: The Standard Deviation function was successfully executed in RStudio and returned numerical values for the sample data.

6. Linear Regression

Linear Regression is used to estimate the trend of the data by fitting a straight line to the values. It uses the relationship between the data points to calculate a trend value for the selected period.

Implementation: linreg.R

Test period: 3

Test Result: The Linear Regression function was successfully executed in RStudio and returned numerical trend values.

7. Relative Strength Index (RSI)

The Relative Strength Index (RSI) is a momentum indicator used to measure the strength and speed of price movements. It helps identify whether the data may be in an overbought or oversold condition.

Implementation: rsi.R

Test period: 3

Test Result: The RSI function was successfully executed in RStudio and returned RSI values for the sample data.

8. Stochastic RSI (StochRSI)

Stochastic RSI applies the Stochastic calculation to RSI values. It shows the position of the current RSI relative to its recent range and can help identify changes in momentum.

Implementation: `stoch_rsi.R`

Test period: 3

Test Result: The StochRSI function was successfully executed in RStudio and returned values for the sample data.

9. Crossover Function

The Crossover function compares two data series and identifies when the first series moves from below the second series to above it. This can be used to identify a possible upward trend signal.

Implementation: `crossover.R`

Test Result: The Crossover function was successfully executed in RStudio and returned Boolean values indicating whether a crossover occurred.

10. Crossunder Function

The Crossunder function compares two data series and identifies when the first series moves from above the second series to below it. This can be used to identify a possible downward trend signal.

Implementation: `crossunder.R`

Test Result: The Crossunder function was successfully executed in RStudio and returned Boolean values indicating whether a crossunder occurred.

11. Conclusion

All nine technical analysis functions were implemented and tested using RStudio. The testing confirmed that the functions could be executed with sample data without producing execution errors. The separate R script files were organized according to the required naming conventions.

12. How to Run the Code

The R scripts can be opened and executed using RStudio. Each indicator is stored in a separate .R file. Open the required R script in RStudio and run the code using the **Run** or **Source** option. The test.R file can be used to test the implemented functions with sample data.